

CHEMISTRY
PHYSICS
GENETICS

YEAR 10

SCIENCE EXAMINATION

SECTION A 43 MULTIPLE CHOICE QUESTIONS (43 MARKS)

SECTION B 11 WRITTEN QUESTIONS (67 MARKS)

- ✓ 1. The valence of the carbonate ion is
A. -3 B. -2 C. -1 D. +1 E. +2.
- ✓ 2. The formula of ammonium phosphate is
A. NH_4P ,
B. $(\text{NH}_4)_2\text{PO}_4$,
C. NH_4P_3 ,
D. $(\text{NH}_4)_3\text{PO}_4$,
E. $\text{NH}_4(\text{PO}_4)_2$.
- x 3. Hydrochloric acid evolves carbon dioxide from which of the following?
A. zinc sulphate
B. copper hydroxide
C. lead nitrate
D. sodium carbonate
E. magnesium chloride
- ✓ 4. An element M has a valence of +3. Another element X has a valence of -2. The formula of the compound formed by combining these two elements is
A. MX ,
B. MX_3 ,
C. M_3X_2 ,
D. M_2X_3 ,
E. M_3X .
- ✓ 5. The element chromium forms the Cr^{3+} ion. The formula for chromium oxide is
A. CrO
B. Cr_2O_3
C. Cr_3O_2
D. CrO_2
E. Cr_2O
- x 6. Acidic oxides react with bases to form
A. an acid and an alkali
B. hydrogen and a salt
C. water and a salt
D. a base and water
E. hydrogen and a carbonate

✓ 7. Which of the following is a non metal:

- (A) Ga (B) Zn (C) Sn (D) Ge (E) Te

X 8 Using the periodic table, the symbol for radium ion, (atomic number of radium = 88) can be predicted as:

- (A) Ra^{-2} (B) Ra^{-1} (C) Ra^{+1} (D) Ra^{+2} (E) Ra^{+3}

X 9 Which of the following substances is insoluble:

- A potassium nitrate B zinc chloride C copper oxide
D sodium acetate E magnesium sulphate

X 10 Which of the following pairs of substances will form a precipitate when mixed:

- A zinc nitrate and copper chloride
B sodium sulphate and magnesium nitrate
C copper nitrate and sodium carbonate
D potassium hydroxide and sodium nitrate
E chromium chloride and calcium nitrate

✓ 11 The formula weight of zinc sulphate is:

- A 129 B 145 C 161 D 97 E 113

X 12 When an element burns in air, the product is known as

- A a salt
B an acid
C a base
D an element
E an oxide

x 13

The electron configuration of an atom with atomic number 19 is:

- (a) 2,8,9
- (b) 2,10,7
- (c) 2,17
- (d) 2,8,8,1
- (e) 2,8,2,7

✓ 14

If an atomic particle has 17 protons, 18 neutrons, and 18 electrons, it is best described as:

- (a) an uncharged atom
- (b) a positive ion
- (c) a negative ion
- (d) a nucleus
- (e) an uncharged ion

✓ 15

A metal M forms a nitrate with the formula MNO_3 . Which of the following formulae for compounds of M is incorrect.

- (a) MCl
- (b) MCO_3
- (c) M_2SO_4
- (d) MOH
- (e) MCH_3COO

x 16

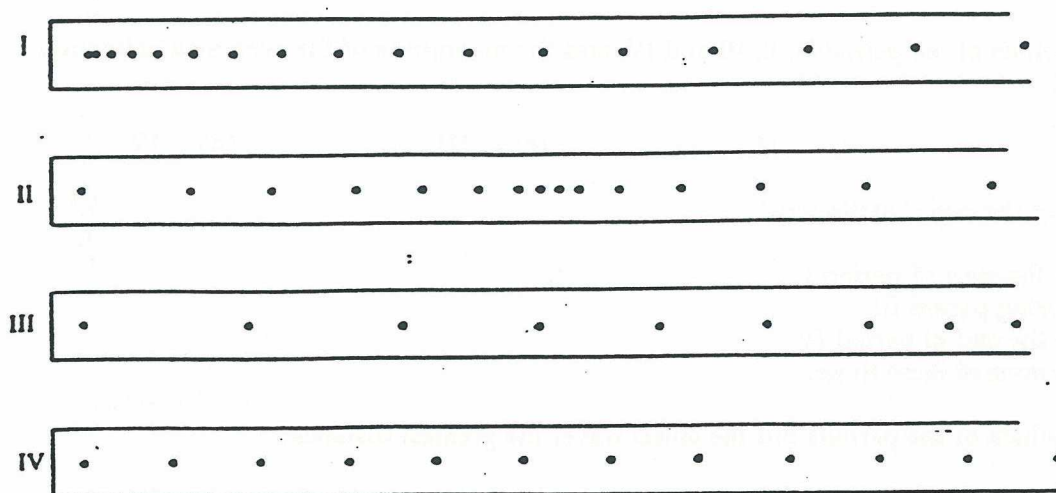
When sulphuric acid is added to copper (II) oxide, one of the products is:

- (a) copper (II) nitrate
- (b) oxygen
- (c) copper (II) hydrogen sulphite
- (d) copper (II) sulphate
- (e) carbon dioxide

- X 17 Hydrochloric acid evolves carbon dioxide from which of the following?
- (A) zinc sulphate
 - (B) copper hydroxide
 - (C) lead nitrate
 - (D) sodium carbonate
 - (E) magnesium chloride
- X 18 Which of the following lists contain all insoluble substances:
- (A) zinc nitrate, copper carbonate, sodium bromide.
 - (B) silver chloride, silver oxide, silver nitrate.
 - (C) copper carbonate, barium sulphate, nickel oxide.
 - (D) potassium carbonate, silver chloride, copper oxide.
 - (E) lead nitrate, iron bromide, sodium hydroxide.
- X 19. A student mixed hafnium benzoate with zinc tannate solution, and noticed a thick white sludge formed. Which of the following is the best conclusion:
(Note: you are not meant to know about the tannate ion or hafnium ion).
- (A) hafnium tannate is insoluble.
 - (B) zinc benzoate is insoluble
 - (C) either hafnium tannate or zinc benzoate is insoluble.
 - (D) both hafnium tannate and zinc benzoate are insoluble.
 - (E) hafnium benzoate does not react with zinc tannate.
- X 20. A precipitate is formed when a concentrated solution of sodium chloride is added to a solution of:
- (a) barium hydroxide
 - (b) sodium sulphate
 - (c) silver nitrate
 - (d) potassium nitrate
 - (e) magnesium chloride
- X 21. A salt can be prepared by:
- (a) adding an organic acid to an alcohol
 - (b) adding copper to a base
 - (c) adding a metal to an acid
 - (d) precipitation when two soluble bases are added together
 - (e) adding sodium chloride to water

22. Which of the following is a scalar quantity?
- (a) velocity (b) acceleration (c) energy (d) force
23. Which one of the following masses is expressed to three significant figures?
- (a) 0.0610 (b) 3.275 (c) 0.00069 (d) 6.370

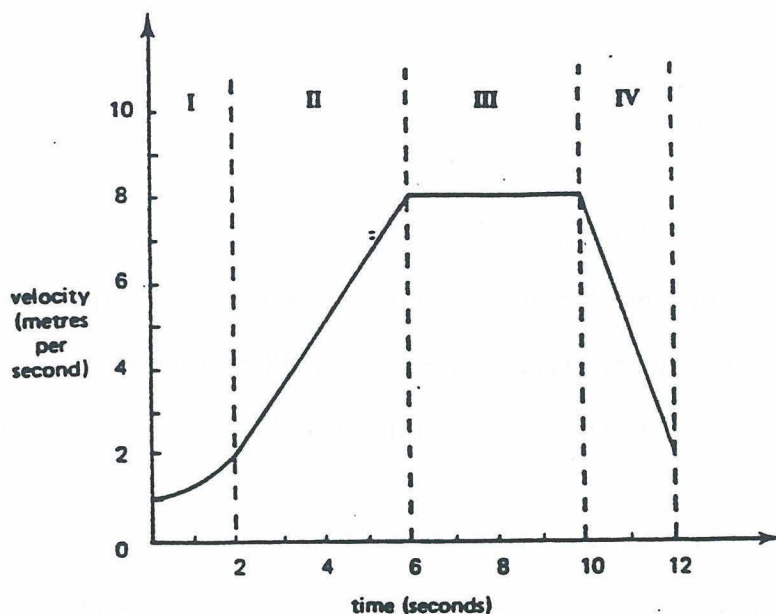
The following diagrams represent the marks made on four separate tapes by the timing device as the cart moves.



24. Which one of the diagrams best indicates that the cart was slowing down throughout the recorded time interval?
- (a) I (b) II (c) III (d) IV
25. Which one of the diagrams best indicates that the cart started from rest and accelerated?
- (a) I (b) II (c) III (d) IV
26. Which one of the diagrams best indicates that the cart was travelling at constant speed?
- (a) I (b) II (c) III (d) IV

THE NEXT 3 QUESTIONS REFER TO THE FOLLOWING INFORMATION.

The velocity-time graph below illustrates the motion of a body moving in a straight line.



✓ 27. During which of the periods I, II, III and IV, was the magnitude of the average acceleration greatest?

- (a) I (b) II (c) III (d) IV

✓ 28. When was the object stationary?

- (a) At the start of period I.
(b) During period III.
(c) At the end of period IV.
(d) At none of these times.

✓ 29. During which of the periods did the object travel the greatest distance?

- (a) I (b) II (c) III (d) IV

✓ 30. A certain organism has a HAPLOID chromosome number of 6. Which statement is CORRECT.

- (a) its muscle cells contain 6 chromosomes
(b) its sperm cells contain 12 chromosomes
(c) its skin cells contain 12 chromosomes
(d) its nerve cells contain 6 chromosomes

✓ 31 Genetics is the study of:

- A reproduction in plants and animals
- B how genes are made up
- C the way characteristics are inherited
- D division of cells
- E sexual reproduction in animals.

✓ 32 Chromosomes:

- A are the real units of heredity
- B contain genes which are the real units of heredity
- C are found in plant cells
- D are found in only animal cells
- E are always visible in cells because they are coloured.

✓ 33 Fertilisation is the name given to:

- A the development of a seed
- B the growth of a new individual
- C the cell division called meiosis
- D the union of two gametes
- E the cell division called mitosis.

✓ 34 In humans, the sex of a child is determined by:

- A the health of the father
- B the health of the mother
- C which sperm does the fertilising
- D which ovum is fertilised
- E when fertilisation occurs.

✓ 35 Identical twins could have:

- A different eye colour
- B different weights
- C different blood groups
- D different shaped noses
- E different sexes.

✓ 36 In meiotic cell division, the:

- A number of gametes is doubled
- B number of chromosomes is doubled
- C number of chromosomes remains the same
- D number of gametes is halved
- E number of chromosomes is halved.

✓ 37 A pure strain or pure breeder is an animal or plant which:

- A has identical genes for the same characteristic
- B has only dominant genes
- C always produces offspring like itself
- D can only breed with another pure breeder
- E has only recessive genes..

✓ 38 A mutation is something which:

- A enables the individual to survive better than before
- B is usually of benefit
- C produces a freak
- D produces a permanent hereditary change
- E occurs at regular intervals.

39. There is a certain degree of variation in the height of a population of human being. This is due to

- 1 mutation
- 2 natural selection
- 3 genetic constitution
- 4 environmental condition

- (a) 3 only
- (b) 1 and 4 only
- (c) 2 and 3 only
- (d) 3 and 4 only

40. In a normal body cell of a dogfish shark there are 24 chromosomes. How many chromosomes are found in each of its gametes?

- (a) 24
- (b) 12
- (c) 8
- (d) 48

41. Which of the following would be an example of vegetative propagation.

- (a) cross pollination
- (b) cuttings
- (c) fertilization
- (d) self pollination

The next 2 questions refer to the following information:

In a certain variety of plant three flower colours are known: red, white, and pink. These flower colours are known to be determined by a single pair of alleles.

42. If heterozygous pink-flowering plants are crossed with homozygous red-flowering plants, which one of the following correctly states the proportions of the offspring by flower colour?

- (a) one half red, one half white
- (b) one half red, one half pink
- (c) one half pink, one half white
- (d) one quarter red, one half pink, one quarter white

43. If homozygous red flowering plants are crossed with homozygous white-flowering plants, which one of the following correctly states the proportions of the offspring by flower colour?

- (a) all pink
- (b) all red
- (c) three quarters red, one quarter white
- (d) one quarter red, one half pink, one quarter white

SHORT-ANSWER QUESTIONS

Answer all questions. Credit will be given for incorrect answers provided your working is clearly and legibly set out.

1. A ticker timer that records 50 dots per second was used to produce the following tape during an experiment. Analyse the tape to determine the acceleration represented.



(5)

2. A toy car is accelerated uniformly from rest over a distance of 135m for 9 seconds

a) Calculate the acceleration.

b) What is the velocity after 6s

(5)

3. In Western movies, crowds of people are sometimes seen firing rifles and hand-guns into the air as a sign of jubilation. Given that a rifle has a muzzle velocity of $3.90 \times 10^2 \text{ ms}^{-1}$ and the bullet is fired straight up, calculate:

(a) the maximum height reached by the bullet.

(b) the time of flight.

(c) the velocity with which it returns to the height from which it is fired.

(5)

4. An aeroplane flies east at 150 km hr^{-1} for 3 hours and then west at 100 km hr^{-1} for 2 hours.

(a) What is the total distance travelled?

(b) What is the average speed?

(c) What is the displacement?

(d) What is the average velocity?

(5)

5.

The world records for some men's track events are listed

Distance(m)	Time(s)	ave speed (ms ⁻¹)
100	9.83	* Do not write here, but on the copy on graph paper
200	19.72	
400	43.86	
800	101.7	
1500	209.5	
5000	778.1	
10000	1620.1	

a) Calculate the average speed for each record. (on the space above) (3)

b) Draw a graph for record speed vs. race distance. (on the paper provided) (3)

c) What does the graph tell you? (2)

6. Classify in tabular form the following substances as metallic element, non-metallic element, ionic compound, covalent compound.
Ne, Na, NO, Co, CO, Cr₂(SO₄)₃, zinc hydroxide, mercury chloride, manganese dioxide, sulfur dioxide, iron, chlorine (3)

7. Classify in tabular form the following substances as soluble or insoluble:
sodium phosphate, barium sulfate, copper hydroxide, potassium carbonate, lead phosphate, zinc chloride. (2)

8. Write balanced equations for:

- (a) zinc and hydrochloric acid
- (b) copper carbonate and nitric acid
- * (c) aluminium hydroxide and sulfuric acid
- (d) magnesium in a solution of silver nitrate
- (e) sodium sulfate and barium nitrate solutions
- (f) calcium chloride solution is mixed with potassium nitrate solution
- (g) Sodium phosphate dissolving in water.
- (h) Lead is dropped into a solution of magnesium sulfate

(15)

9. When a certain strain of yellow guinea pigs are crossed with white guinea pigs, cream offspring are produced. This is a case of incomplete dominance.

Using the symbols Y for yellow colour and W for white colour, give the genotypes and phenotypes for the following crosses.

- (a) Two cream guinea pigs
(b) A cream guinea pig and a white guinea pig.

(5)

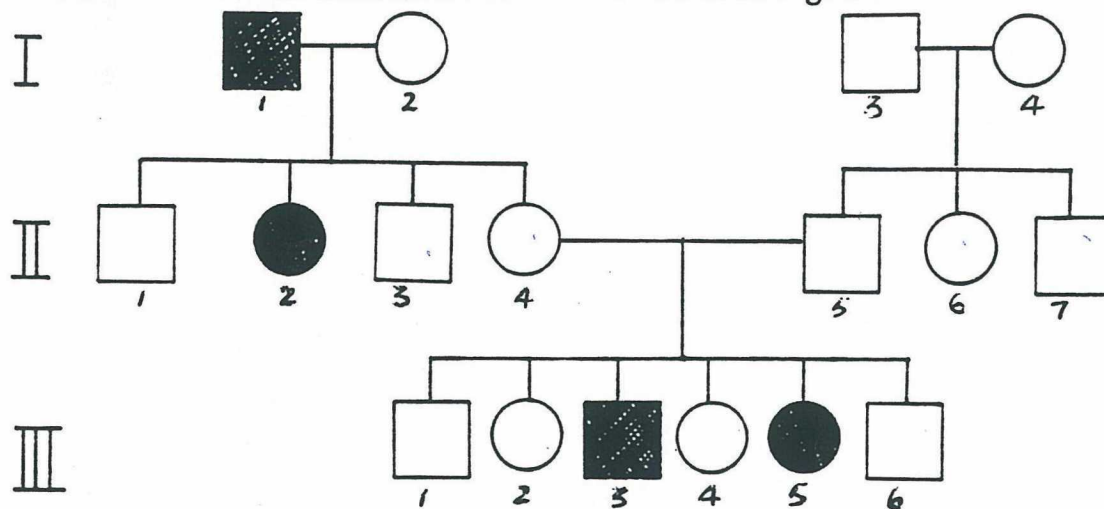
10. Colour blindness is a sex linked recessive disorder.

- (a) Explain why, in a family of 2 boys and 2 girls, both boys are colour blind, but neither the girls nor their parents are colour blind.
(b) Draw a pedigree (family tree).

(3)

(2)

11. An inherited condition known as cataract is found in canaries. The pedigree below shows the occurrence of this condition in 3 generations.



- (a) Is cataract dominant or recessive? Why?
(b) Determine the genotypes I 2, I 3, II 4, III 5.
(Use the symbols N and n)
(c) If individuals II 1 and II 2 were mated, what would be the genotypes, phenotypes and proportions of the offspring.

(2)

(4)

(3)

CHEMISTRY 4

1. In 1981, four groups of chemists measure the atomic weight of an exciting new element, Wilsonium. Their results after months of research are tabulated below:

Group 1		Group 2		Group 3		Group 4	
Trial	At.Wt	Trial	At.Wt	Trial	At.Wt	Trial	At.Wt
1	253	1	207	1	241	1	229
2	207	2	214	2	239	2	273
3	261	3	218	3	221	3	239
4	217	4	216	4	235	4	285
5	285	5	212	5	228	5	251
6	240	6	215	6	229	6	259
Av. = 243		Av. = 214		Av. = 232		Av. = 255	

In 1989, a new method reveals the true atomic weight of Wilsonium is 241.

- (a) Which group had greatest precision, and why? (2)
 - (b) Which group was most accurate? (1)
 - (c) Use the terms high, medium and low to describe both the accuracy and precision of group 4. (2)

2. Define:
 - (a) molecular weight
 - (b) the mole (2)

- 3 A 27.3g sample of potassium phosphate is reacted with excess calcium chloride:

$$2 \text{K}_3\text{PO}_4 + 3 \text{CaCl}_2 \rightarrow \text{Ca}_3(\text{PO}_4)_2 + 6 \text{KCl}$$
 - (a) What mass of potassium chloride is produced? (4)
 - (b) What mass of calcium chloride is used up? (2)
 - (c) What mass of calcium phosphate is produced? (2)

- 4 Pentane burns in air according to the following equation:

$$\text{C}_5\text{H}_{12} + 8\text{O}_2 \rightarrow 5\text{CO}_2 + 6\text{H}_2\text{O}$$
 - (a) What volume of CO_2 is produced when 1.5kg of pentane (C_5H_{12}) is burnt? (3)
 - (b) What mass of oxygen is consumed when 207L of CO_2 is produced? (3)

5. (a) 5L of a gas at a temperature of 50°C and a pressure of 75kPa, occupies what volume at 50°C and a pressure of 102kPa. (3)

